

Design and Implementation of a Custom Low-Latency Wireless Control Link for UAV Systems

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1 Introduction

1.1 Motivation

Reliable telecommand and telemetry are fundamental for UAV operation, yet many commercial solutions rely on proprietary technologies or do not provide flexible configurations. LoRa is a wireless digital radio technology used mainly for IoT devices, preferred for low power consumption and long range, interference resistant communication. Because the transmitted packets are small, LoRa may be suitable despite its low data rate.

1.2 Scope of Project

This work covers the design and assembly of a custom PCB integrating an STM32G474 microcontroller and an Ebyte E22-900M22S LoRa module, intended as a building block for a UAV command and control link. The firmware implements the Semtech LoRa driver, a minimal link layer protocol, and analog joystick input measurement. The transmitter side of the link was brought up and verified. The transmitted signal was captured and characterised using an RTL-SDR, confirming correct frequency, occupied bandwidth, and continuous operation at the configured 50 Hz packet rate. The receiver board, range testing, and end-to-end link evaluation are outside the scope of this work and are identified as directions for future development.

2 Related Work

2.1 LoRa in UAV Command and Control Links

ELRS represents the state of the art in open source UAV control links, achieving up to 200 Hz packet rate at 868 MHz using LoRa with small payloads and low SF, trading receiver sensitivity for lower latency or 1000 Hz at 2.4 GHz. The present work targets 50 Hz, prioritizing link budget over update rate, which is sufficient for UAV command and control applications where latency below 20 ms is not a hard requirement. [1]

3 Hardware Design

3.1 System Architecture

The system consists of two boards, a designated transmitter and receiver. They communicate with a PC and can be powered or charged via USB. The boards can also be operated from a battery. The transmitter has left and right joysticks connected which are measured with ADC then embedded in the payload. The radio is managed over SPI commands. The payload is written in the transmit buffer, then transmitted over the air. The receiver decodes the packet, then outputs PWM signals on the four channels.

3.2 PCB Design: Schematic and Layout

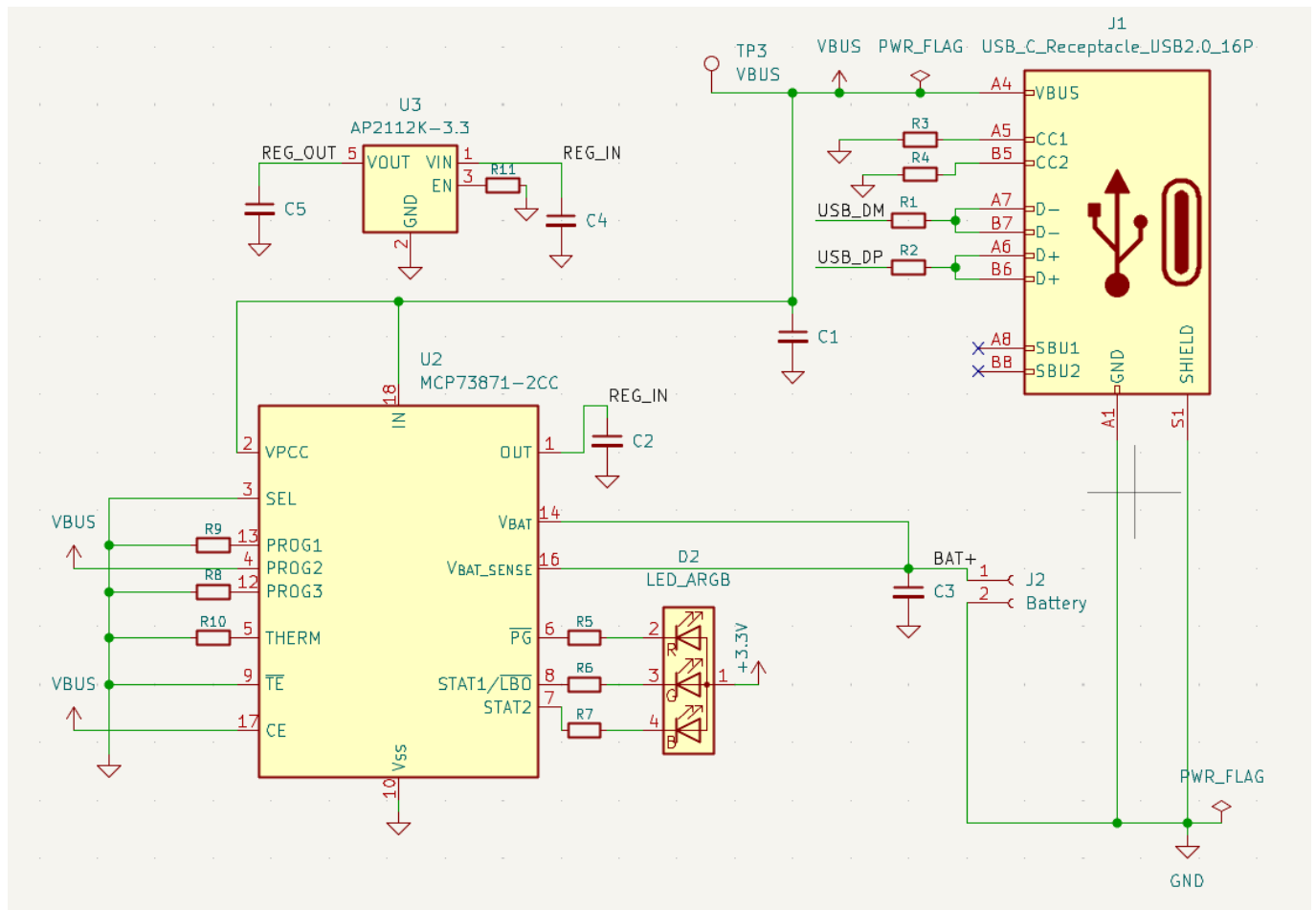


Figure 1: Power section schematic.

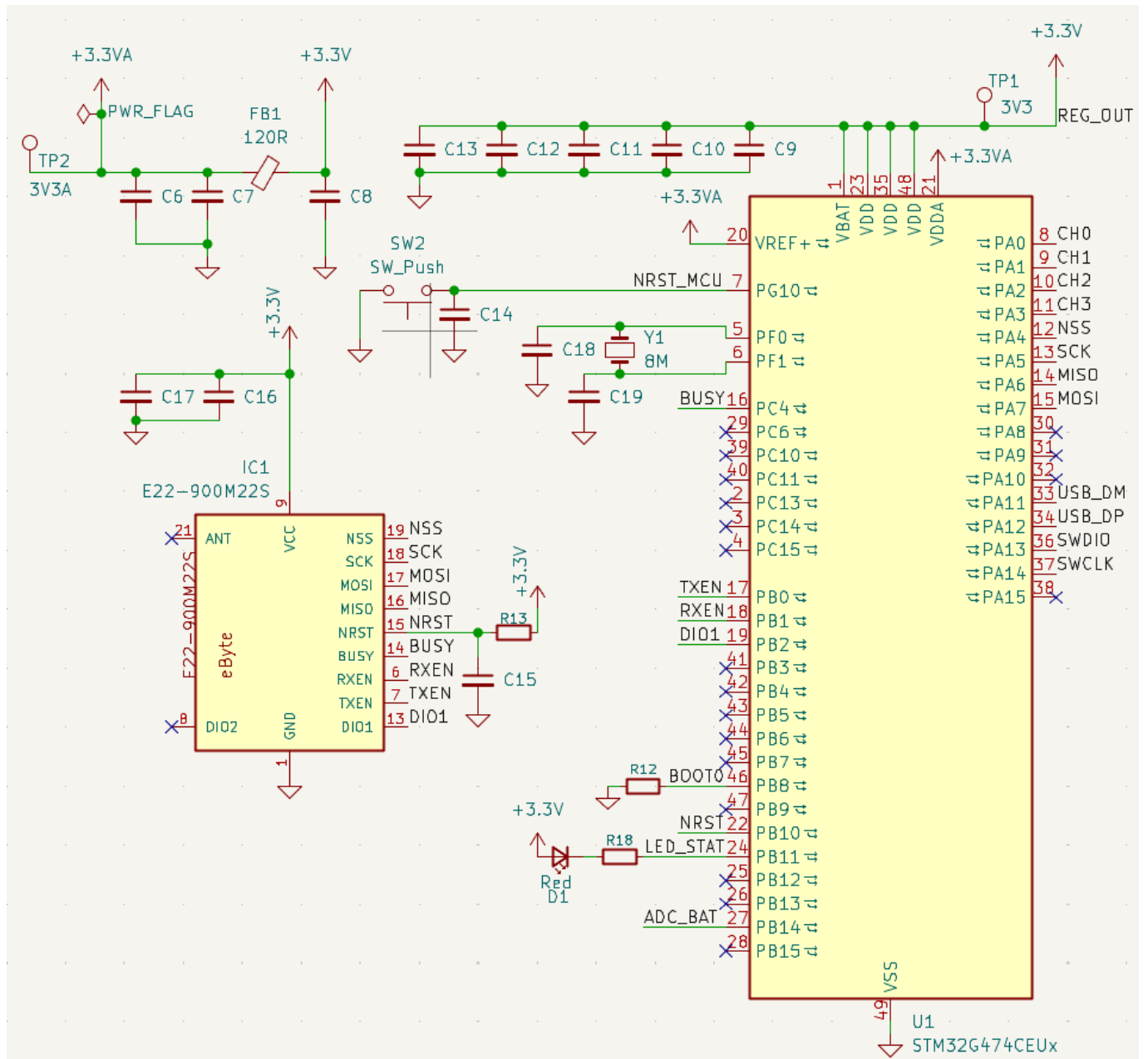


Figure 2: MCU and Radio schematic.

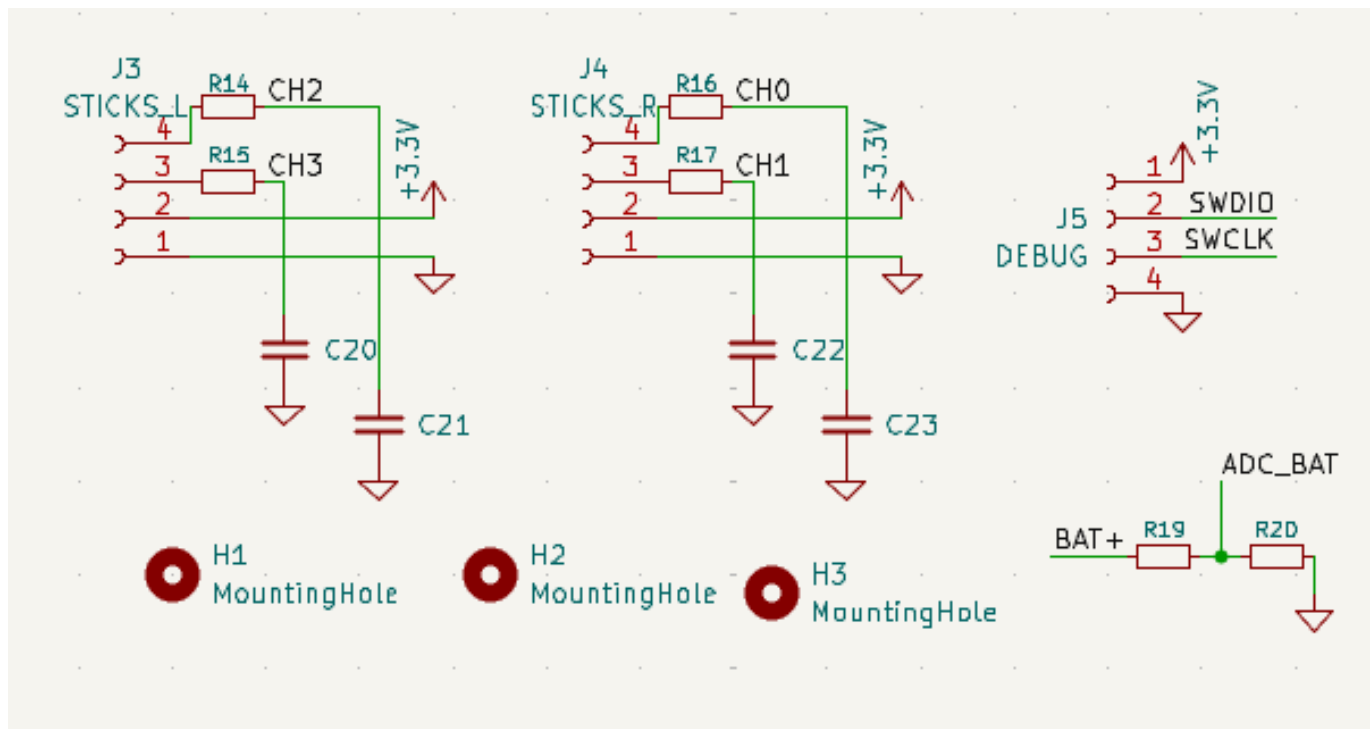


Figure 3: Connectors schematic.

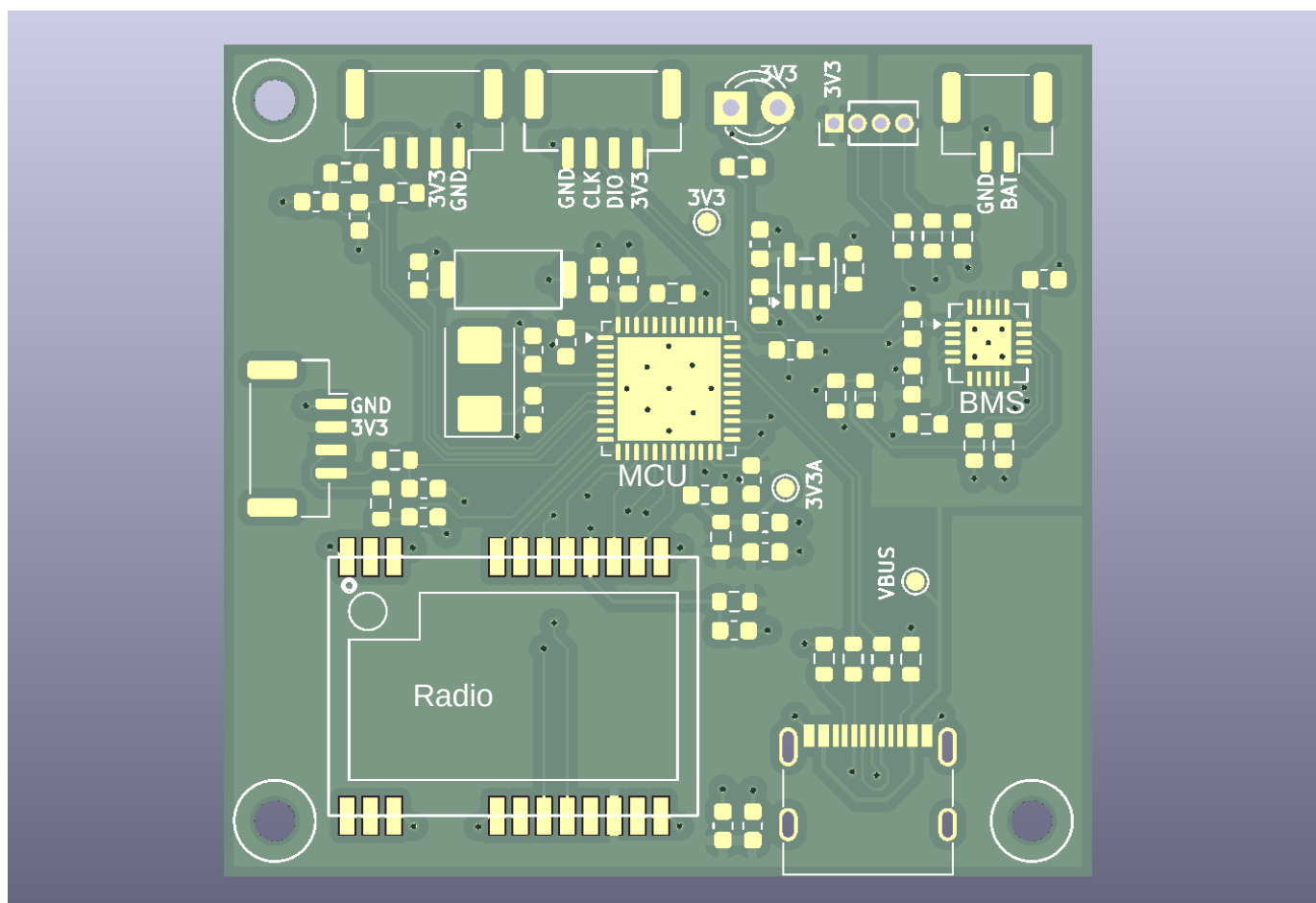


Figure 4: PCB layout.

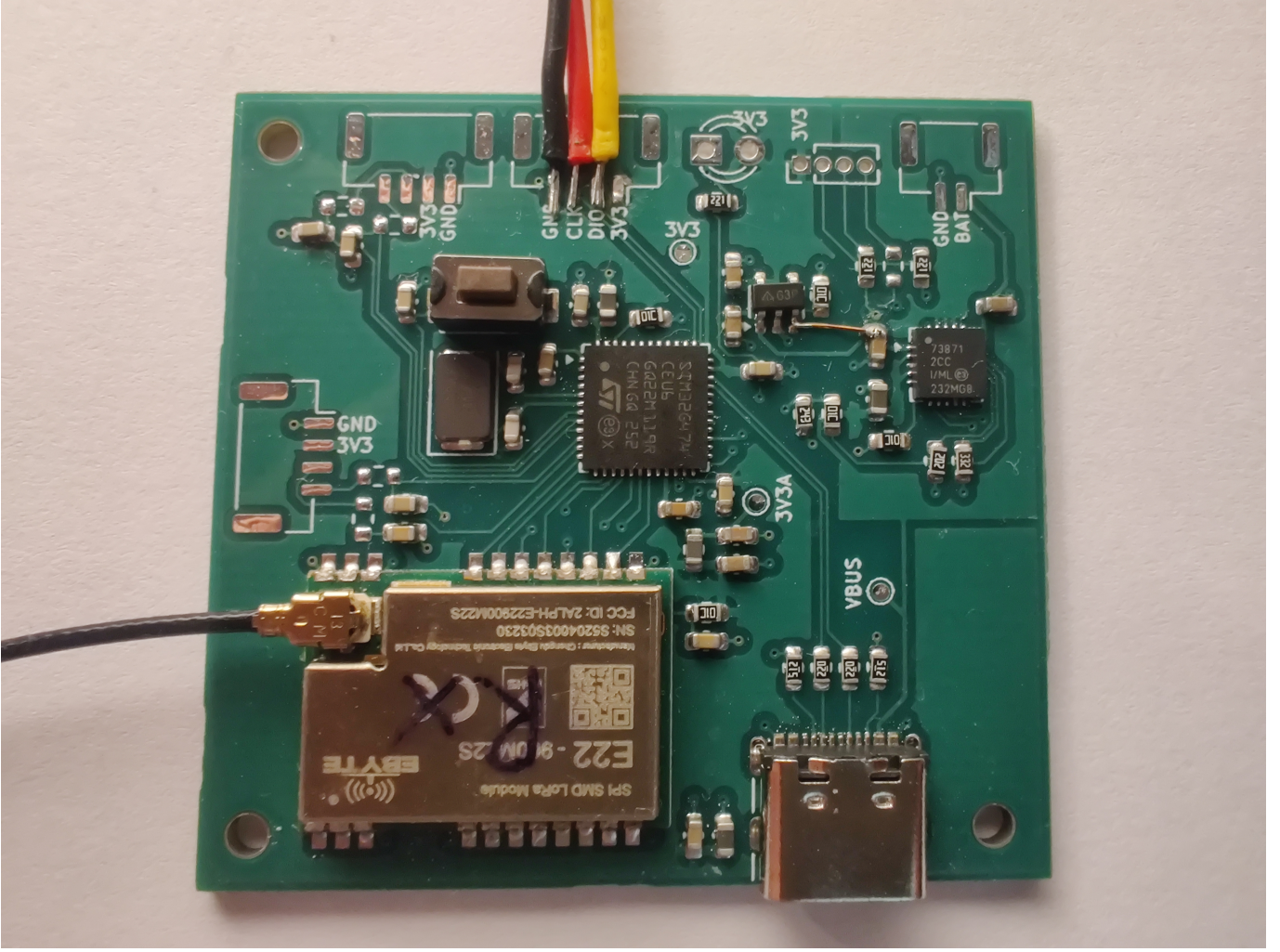


Figure 5: Assembled board.

The board is a 2-layer FR4 design with 47.0 mm * 45.0 mm dimensions and 3 M2 mounting holes. The connectors are 1.25 mm JST SH. The 3V3A analog supply is separated from the digital 3V3 by a 120 Ω ferrite bead and capacitors, reducing switching noise interfering with ADC measurements. Decoupling capacitors are placed at each IC's power pins. The radio is oriented such, that the high speed SPI traces are as short as possible. The 8 MHz crystal is positioned close to the MCU with the appropriate load capacitors. The USB traces are almost length matched with 1 mm difference, which is compliant with 2.0 FS standard.

3.3 Component Selection Rationale

The board is powered via a USB-C receptacle. An MCP73871-2CC [2] battery charger manages the charging of a single-cell lithium battery from VBUS with a charge current set via resistors on the PROG pins while simultaneously feeding an AP2112K-3.3 LDO [3] that produces the 3.3V system voltage. The LDO was chosen because it can supply 600 mA as the radio requires large currents. Without VBUS the system receives power only from the battery. An RGB LED indicates the charger status via the STAT1 and STAT2 outputs of the MCP73871. CC resistors on the USB-C connector configure it as a USB 2.0 device that accepts 5V bus power.

The MCU is an STM32G474CEU6 [4], which has a Cortex M4 core running at 170MHz. It was preferred because of high clock speed, equipped with hardware floating point unit and 12 bit ADC running as high as 4 MSPS. Certain pins are routed to connectors for easy access, such as debug and the 4 analog input channels for UAV control, which can be configured as PWM output so the transmitter and receiver boards can be identical. The radio is an Ebyte E22-900M2S, which

includes an SX1262 LoRa chip [5] alongside an RF frontend with a uFL connector for the antenna and a TCXO. The radio operates at the 868MHz ISM band.

3.4 Design for Low Latency: Key Trade-offs

LoRa introduces a configurable tradeoff between data rate and receiver sensitivity mainly by selecting spreading factor, bandwidth. SX1262 supports SF5 .. SF12 and bandwidths 8 kHz .. 800 kHz. A good compromise is SF6 with 125 kHz bandwidth giving a symbol time of 0.51 ms and effective data rate of 9.375 kbps. For a 6 byte payload and LoRa overhead the time on air is 18ms resulting in a comfortable 50 Hz packet rate. Note that EU regulation only allows 10 % duty cycle at most, so this configuration is not compliant. [6] [7]

4 Firmware and Driver Integration

4.1 LoRa Driver Stack and Configuration

The C driver is supplied by Semtech, with regular updates. The user must implement the hardware specific low level functions such as SPI transactions, radio reset or external interrupt handling.

4.2 Link Layer Protocol and Packet Structure

The payload is 6 bytes, consisting of 4 ADC channels, each mapped to an 8 bit value with the following function:

```
uint8_t x8 = (uint8_t)round((x12 * (255.0 / 4095.0)));
```

The transmitter sends packets at 50 Hz without receiver acknowledge. After a preset amount of time the transmitter requests telemetry by setting a flag in the payload and goes into RX mode for 50 ms. Telemetry is transmitted as best effort, if the transmitter didn't get the packet, it will ask again at the next period. This is superior to shifting by 4, because the average error is halved. The other 16 bits are used in arbitrary configuration. The telemetry packets include 1 byte RSSI and SNR calculated by the receiver. A 16 bit received packet counter helps to measure packet loss. Battery voltage is included on 12 bits and the last 4 bits are used for arbitrary flags.

5 Signal Acquisition and Analysis

5.1 RTL-SDR Setup and IQ Capture Pipeline

The SDR used for capture is an RTL_SDR v4 equipped with a TCXO. The capture pipeline was constructed in GNU Radio Companion, consisting ofn RTL_SDR Source connected to a Waterfall and a Frequency Sink. The RF gain is set to 0 dB to avoid saturating the ADC while AGC was turned off. The center frequency is 868.1 MHz, same as the radio setting. The sample rate should be at least 125 kHz because of complex sampling plus a small headroom for the filter cut-off range on the Frequency and Waterfall Sinks so 256 KHz is more than enough.

5.2 Spectral Analysis and Signal Characterisation

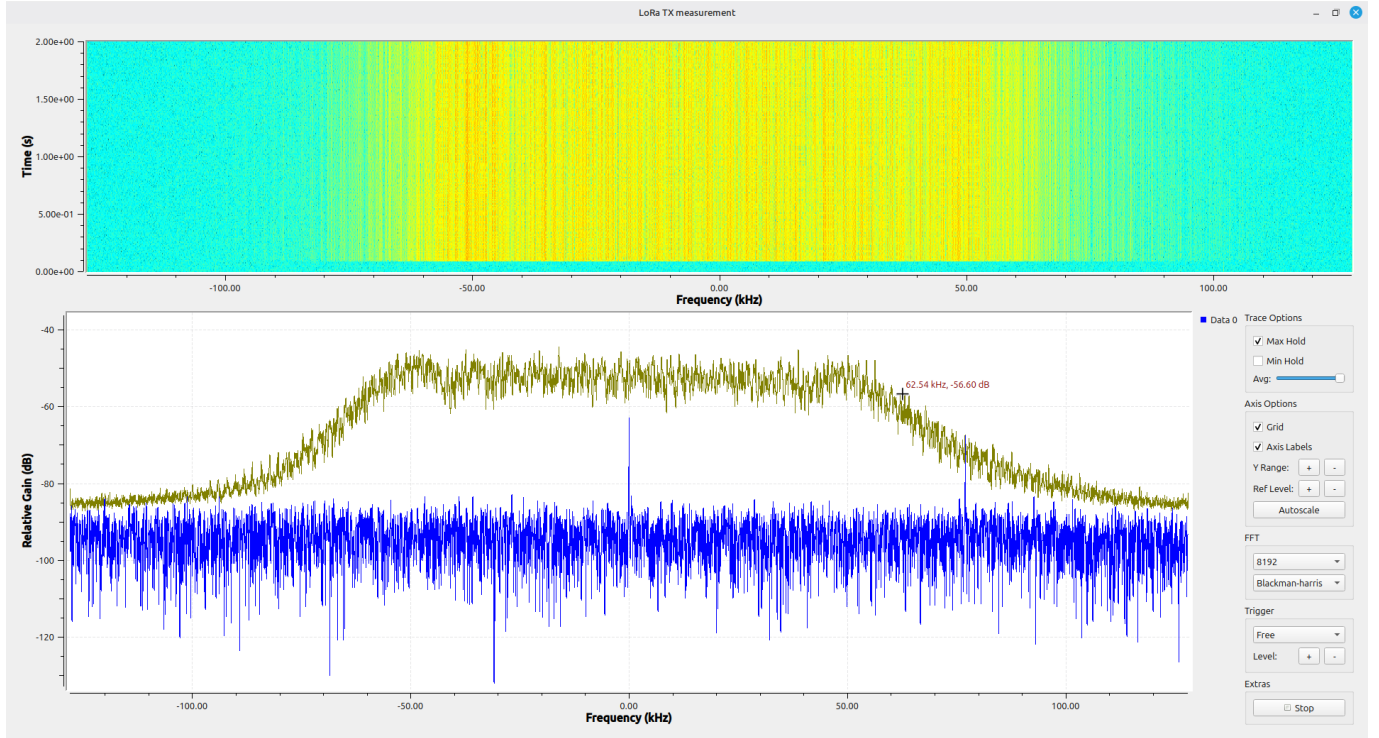


Figure 6: Transmitted signal in frequency domain.

The signal is well contained within ± 62.5 kHz of the carrier and centered around 0 Hz indicating no drift or Doppler-shift which is expected of the TCXO's and the stationary boards. The 50 Hz packet rate is not visible possibly due to the waterfall's resolution, but it's clear that the transmitter is almost always on.

6 Conclusion and Future Work

This work presented the design, assembly and partial validation of a custom PCB for a LoRa based UAV command and control link. The receiver board, end-to-end link testing, range measurements, and packet loss characterization remain as future work. A natural next step is writing the receiver firmware, after which round-trip latency and RSSI versus distance measurements can be conducted to fully evaluate the link against UAV control requirements. Regulatory compliance should also be addressed in a deployed configuration, either by reducing the packet rate or implementing listen before talk.

References

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